

■ 1.8.10 Contour Plots

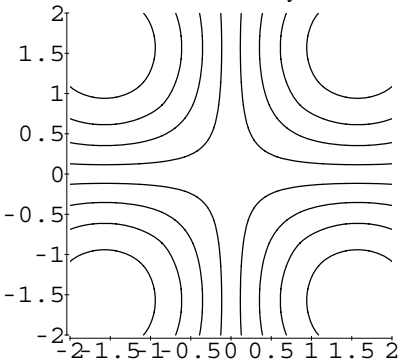
When you are trying to understand the form of a particular surface, it is often useful to look at the surface in several different ways. `Plot3D` gives you a realistic three-dimensional picture of the surface. You can use `ContourPlot` to get a “topographic map” of the surface.

```
ContourPlot[f, {x, xmin, xmax}, {y, ymin, ymax}]
```

make a contour plot of f

Contour plots.

This gives a contour plot of the function $\sin(x)\sin(y)$.
`In[1]:= ContourPlot[Sin[x] Sin[y], {x, -2, 2}, {y, -2, 2}]`



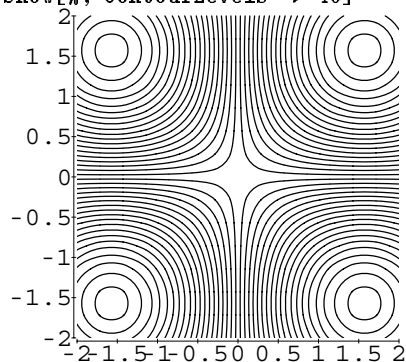
The “contour lines” are just like those in a standard topographical map. They join points on the surface that have the same height. The contour lines correspond to an evenly-spaced sequence of z values.

option name	default value	
PlotPoints	15	number of evaluation points in each direction
PlotRange	Automatic	the range of values to be included; you can specify {zmin, zmax}, All, etc.
ContourLevels	10	number of contour lines between zmin and zmax
ContourSpacing	Automatic	spacing in z between successive contour lines

Some options for ContourPlot.

This increases the density of contour lines shown.

```
In[2]:= Show[%, ContourLevels -> 40]
```



If you do not evaluate your function on a fine enough grid, there may be inaccuracies in your contour plot. One point to realize is that irregularities become obvious in `Plot3D` and `ContourPlot` in somewhat complementary circumstances. `Plot3D` gives an irregular surface if your function varies too rapidly. `ContourPlot`, on the other hand, gives a regular pattern of contour lines when your function varies rapidly, but can give irregular contour lines when the function is almost flat.